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Semiconductor-metal transition in thin VO₂ films grown by ozone based atomic layer deposition

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Thin films of vanadium dioxide (VO₂) have been grown by a low temperature atomic layer deposition process at 150 °C using tetrakis[ethylmethylamino]vanadium as a vanadium source and ozone as reactant gas. Films deposited on SiO₂ were amorphous, but during a thermal treatment at 450 °C tetragonal VO₂(R) was formed. During *in situ* x-ray diffraction measurements, the semiconductor–metal transition was observed as a reversible transition between VO₂(M1) and VO₂(R) near 67 °C. Correlated with this phase change, a reversible change in resistivity was observed of more than two orders of magnitude for a film of 42 nm thickness. © 2011 American Institute of Physics. [doi:10.1063/1.3579195]

In the large group of vanadium oxides, VO₂ has the interesting feature that a large reversible change in resistivity occurs near a temperature of 68 °C,¹ which is referred to as a metal–insulator transition or semiconductor–metal transition (SMT). The transition is known to be accompanied with a transformation in crystallographic structure, from monoclinic VO₂(M1) below the transition temperature to tetragonal VO₂(R) above.² For single crystals the resistivity change can be as high as five orders of magnitude and occurs in a very narrow temperature range of only a few degrees Celsius.³ This property makes VO₂ an interesting candidate for applications in resistive memories and switches in microelectronics. In that case one is interested in inducing the resistive switch electrically rather than thermally and several groups succeeded already in making such test devices.^{4,5}

A lot of results are reported in literature on thin film growth of VO₂, typically using pulsed laser deposition,⁶ reactive sputtering,^{7,8} or chemical vapor deposition^{9–11} as deposition techniques. For applications in microelectronic devices, however, atomic layer deposition (ALD) is gaining importance for depositing oxide films, due to aggressive scaling of device dimensions and increasing integration complexity, requiring in many cases conformal depositions in three-dimensional structures. The key feature for this technique is the self-limiting character of the surface reactions, making layer by layer growth possible.^{12,13} The main advantages are the excellent control of thickness and stoichiometry and a very good uniformity and conformality on complex structures. Also the typical deposition temperatures are rather low compared to other deposition techniques. However, ALD processes for vanadium oxides mostly result in V₂O₅ growth.¹⁴

In this work, it is shown that amorphous VO₂ films can be deposited with an ALD process at only 150 °C. During an additional thermal treatment at 450 °C in inert gas VO₂(R) is observed as the first and only crystalline phase, which on cooling down transforms to VO₂(M1). A SMT is observed

with a resistivity change of more than two orders of magnitude for a film with a thickness of 42 nm.

Thin vanadium oxide films were grown in an experimental low-pressure ALD reactor^{15,16} on Si substrates with 100 nm thermally grown SiO₂. Tetrakis[ethylmethylamino]vanadium {V(NEtMe)₄}, supplied by Air Liquide and denoted here as TEMAV, was used as metal-organic vanadium precursor, which has an oxidation state of 4+ for the vanadium, in contrast to other vanadium precursors like VO[O(C₃H₇)ⁱ]₃ where it is 5+. This could possibly facilitate the formation of VO₂ instead of V₂O₅, since 4+ is also the oxidation state of vanadium in VO₂. As reactant gasses, both H₂O (thermal), O₂ (plasma enhanced) and O₃ were tested. Although it seemed possible to obtain similar results with all three reactants, the process with O₃ behaved as the most reliable. Therefore, only results of the ozone based ALD process will be discussed in detail in this letter. The ozone was produced from a pure flow of O₂ with an OzoneLab™ OL100, resulting in an O₃ concentration of 175 μg/ml. The base pressure in the reactor was 10^{−6} mbar, whereas the partial pressures for TEMAV and ozone were 10^{−4} mbar and 5 × 10^{−1} mbar, respectively. Ar was used as a carrier gas for the TEMAV. Since the main characteristic of ALD is the self-limiting nature of the surface reactions, it is important to verify in which temperature range this behavior is observed. Therefore, a process was performed at several substrate temperatures, consisting of 200 times 2 s exposure with TEMAV while no reactant gas was used. At 200 °C, a film was grown with a growth rate of 0.3 Å/s of TEMAV exposure as derived from an x-ray reflectivity (XRR) measurement, corresponding to thermal decomposition of the precursor. While no clearly detectable layer was observed with XRR when the temperature was lowered to 175 °C, x-ray fluorescence (XRF) revealed that there was still some vanadium deposited. The XRF signal ratio V/Si was ten times lower than the deposition at 200 °C. Although very low, this signal was again lowered five times at 150 °C substrate temperature. These results suggest that the maximum ALD temperature for TEMAV is 150 °C, although at 175 °C the decomposition of TEMAV is actually very small. Saturation experi-

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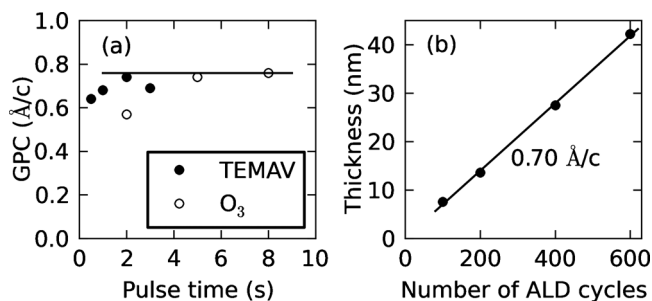


FIG. 1. ALD characteristics for the process TEMAV—O₃ at 150 °C: (a) Growth per cycle vs exposure times for TEMAV with a fixed O₃ exposure of 5 s (filled symbols) and for O₃ with a fixed TEMAV exposure of 2 s (open symbols); (b) Thickness of the deposited films as a function of the number of ALD cycles.

ments for the ALD process were performed at 150 °C and it was concluded that 2 s TEMAV and 5 s ozone resulted in a saturated ALD growth [Fig. 1(a)]. Pump times were, respectively, 25 s and 15 s after the TEMAV and ozone pulse. The corresponding growth per cycle (GPC), which is derived from dividing the measured thickness of the film by the number of ALD cycles (100 cycles in these cases), is 0.76 Å/c. Although a slight decrease in GPC is observed for 3 s TEMAV with respect to 2 s TEMAV exposure, it is within the error range of the XRR measurements. The uniformity of the saturated process at 150 °C was excellent, with a deviation in thickness of less than 2% on a sample of 30 cm². Also the linearity of the ALD process was good, as shown in Fig. 1(b). From the slope of a linear fit a GPC of 0.7 Å/c was extracted.

In order to determine the stoichiometry of the films, x-ray photoelectron spectroscopy (XPS) measurements were performed on (i) a film grown with the O₃-based ALD process and (ii) a film grown with an O₂-plasma enhanced ALD process, having a long plasma exposure (Fig. 2). For the latter it seemed that, as long as the plasma exposure was sufficient, crystalline V₂O₅ could be observed with x-ray diffraction (XRD), as shown in Fig. 3. The O 1s peak (V–O bond) was used as a reference for the binding energy (530.0 eV). For the O₂ plasma grown film, the binding energy of the V2p_{3/2} peak was fitted as 517.2 eV, while for the ozone-based ALD film this value was 515.95 eV. According to Silversmit *et al.*,¹⁷ these values are related to the V⁵⁺ and V⁴⁺ oxidation states, respectively, i.e., to V₂O₅ and VO₂, respec-

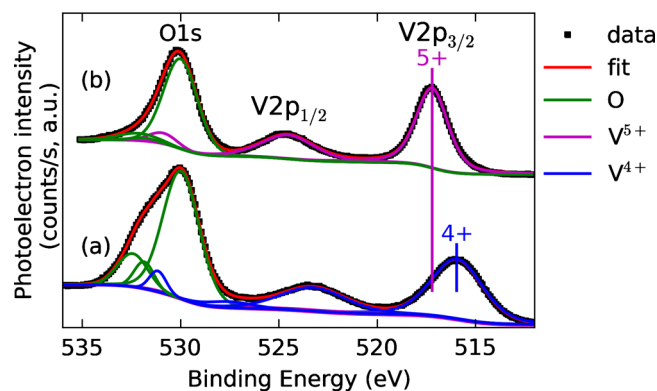


FIG. 2. (Color online) XPS of (a) an as deposited O₃-grown ALD film, and (b) a film grown by an O₂-plasma enhanced ALD process with long plasma exposure.

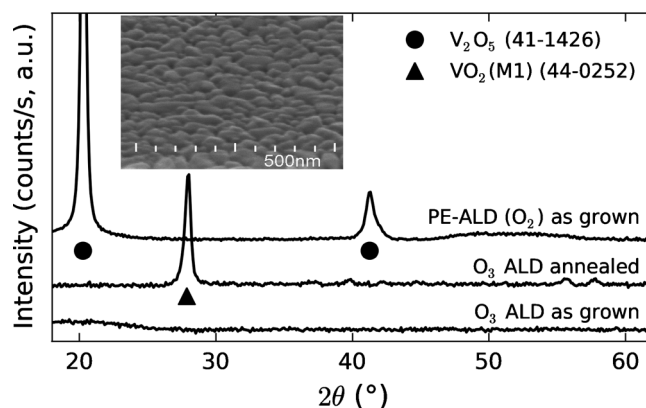


FIG. 3. XRD spectra of an O₃-grown ALD films, the same film annealed for 30 min in He at 450 °C and a film grown by an O₂-plasma enhanced ALD process with long plasma exposure. In the inset, a SEM picture is shown for the annealed O₃-grown ALD film.

tively. Thus, it can be concluded that the O₃-based ALD films were VO₂ as deposited. Additionally, no detectable amount of impurities like C and N were observed in the as grown films.

From XRD ($\lambda=0.154$ nm) of the as deposited films it was concluded that O₃-grown films, independent of the thickness, showed no sign of crystallinity (Fig. 3). Therefore, on some films isothermal anneals were performed at different temperatures and for varying times in an inert atmosphere (He). These thermal treatments were performed in a experimental heating chamber which was placed in a Bruker D8 Discover XRD system.¹⁸ The XRD pattern was captured every 5 s with a linear Vantec detector, having a range in 2θ of 20°. For the 42 nm film it was observed that at 450 °C in He the crystallization started within less than 10 min, while the integrated peak intensity slowly increased during the first 30 min (Fig. 4). Although there are many possible crystalline phases of VO₂, the position of the peak ($27.65^\circ \pm 0.05^\circ$ as 2θ value) suggests that it corresponded with the 110 plane of tetragonal VO₂(R) [Joint Committee for Powder Diffraction Standard (JCPDS) 44-0253], while no signs of other vanadium oxide phases were observed. Scanning electron microscopy (SEM) of the annealed film of 42 nm (inset of Fig. 3) revealed a continuous polycrystalline film, while from atomic force microscopy (AFM) the root mean square value for the surface roughness was derived as 4.5 nm. For thinner films the roughness increased substantially when the same

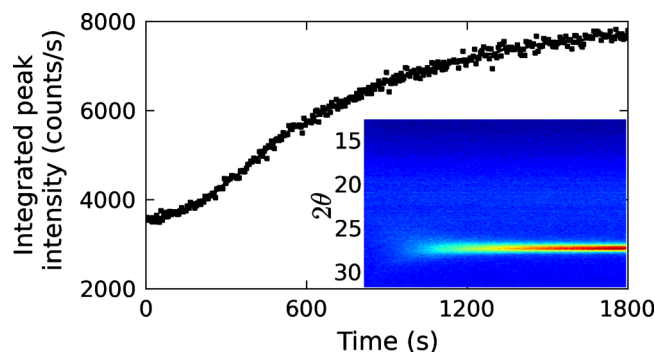


FIG. 4. (Color online) Integrated intensity for the diffraction peak of the VO₂(R) 110 plane during an isothermal anneal of 30 min at 450 °C in He for a film of 42 nm. The inset shows the *in situ* XRD measurement from which the peak intensity was integrated.

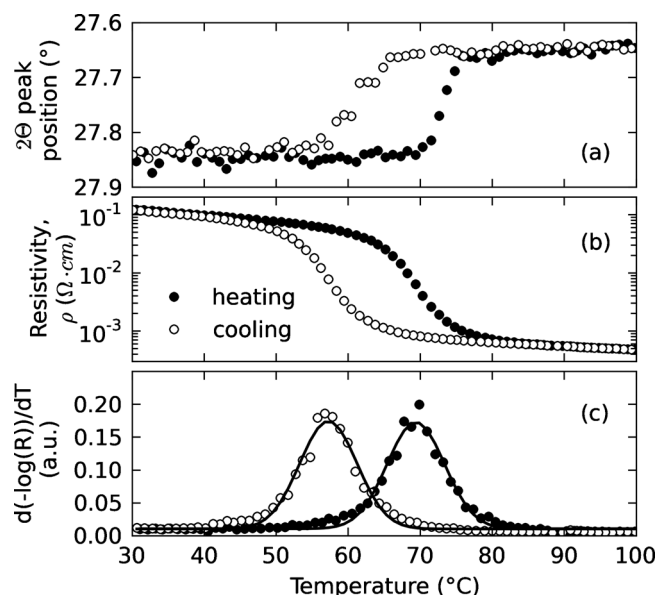


FIG. 5. SMT as observed during a thermal cycle between 30 and 100 °C, with fixed heating and cooling rate of 0.1 °C/s: (a) *in situ* XRD, showing the change in crystal structure, (b) the correlated change in resistivity, and (c) a Gaussian curve fit to $d[-\log(r)]/dT$ for the analysis of the resistivity.

heating procedure was performed. More evidence for the assumption of $\text{VO}_2(\text{R})$ is provided by the observation that the diffracted peak position shifted between 27.85° and 27.65° when the sample temperature was varied between 30 and 100 °C [Fig. 5(a)]. This corresponded with the reversible transformation between $\text{VO}_2(\text{M1})$ (JCPDS 44-0252) at low temperature and $\text{VO}_2(\text{R})$ at high temperature. In addition, Fig. 3 also shows the XRD pattern for the annealed film, measured at room temperature. The transition temperature was approximately 73 °C during heating and 61 °C during cooling. Thus, the mean transition temperature was 67 °C and the hysteresis 12 °C.

In the same heating stage as used for the *in situ* XRD measurements 2 point resistivity measurements were performed with a Keithley 2000 Multimeter in order to correlate the crystallographic phase transformation with a change in resistivity. Since the measured resistivities were in the range of kilo-ohms or higher, the contact resistance was negligible and the use of a 2 probe setup was justified. In addition, 4 point resistivity measurements were performed at room temperature in order to obtain absolute resistivity values. Figure 5(b) shows the resistivity as a function of temperature for the 42 nm film crystallized by annealing three times 10 min at 450 °C in He. A SMT is clearly observed with a transition temperature a few degrees Celsius lower than the crystallographic phase transformation and a resistivity ratio of approximately 200 in the range 30–100 °C. Transition temperatures were derived by fitting a Gaussian curve to

$d[-\log(R)]/dT$ for both heating and cooling [Fig. 5(c)]. This resulted in a mean transition temperature of 63 °C, which is 4 °C lower than derived from *in situ* XRD, and a hysteresis of 12 °C.

In conclusion, ozone based ALD was used to deposit thin films of vanadium oxide, with TEMAV as a vanadium precursor. XPS revealed that the stoichiometry was VO_2 . Since all films were amorphous, isothermal anneals were applied in an inert atmosphere, resulting in tetragonal $\text{VO}_2(\text{R})$. *In situ* XRD measurements also showed a reversible transformation between $\text{VO}_2(\text{M1})$ and $\text{VO}_2(\text{R})$ near 67 °C with a hysteresis of approximately 12 °C. This crystallographic phase transformation was correlated with a SMT as observed during resistivity measurements. The mean transition temperature in this case has been derived as 63 °C, with a hysteresis of 12 °C. The resistivity dropped by a factor of 200 in the range 30–100 °C for a 42 nm film.

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